

Robust Charging Infrastructure Design under Demand Uncertainty

A solar-powered charging station for a shared e-micromobility fleet

Which station do you build when the future is uncertain?

Theme 3 — solar charging station

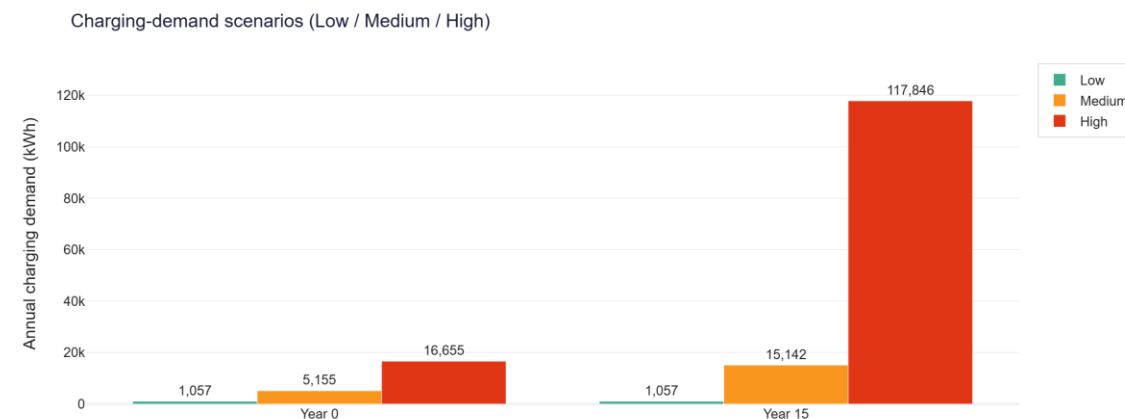
Theme 2 — charge/discharge profiles

The problem: sized for average, or for the worst?

The dilemma. Size for average demand and the fleet is stranded at peaks; size for the worst case and capital is wasted.

A constrained grid. Upgrading the connection needs a costly network reinforcement, so the evening peak must come from on-site solar + storage.

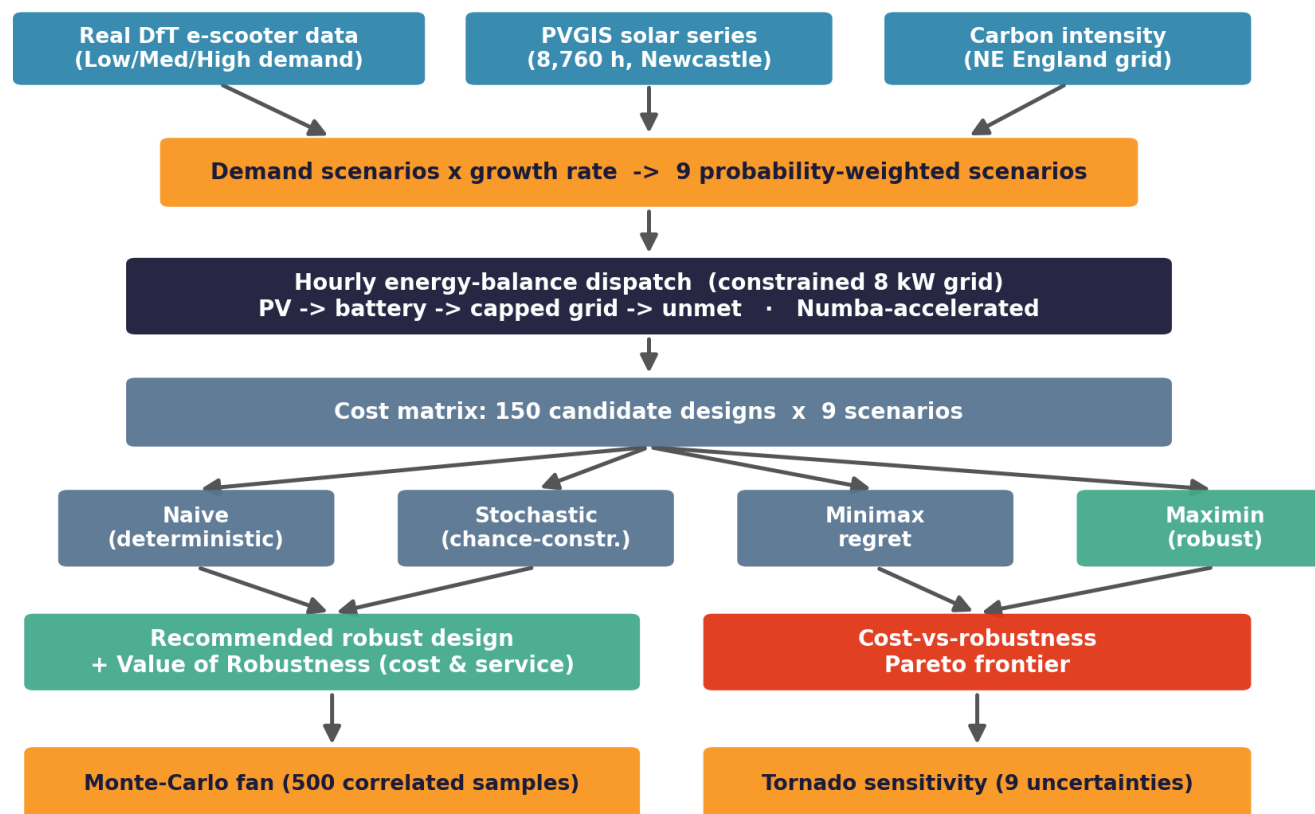
Grounded in real data. Low / Medium / High scenarios are built from the national DfT e-scooter trial dataset — real trips, fleet, utilisation, distance.



Demand scenarios derived from real trial data.

Approach — a reproducible robust-design pipeline

Methodology — Robust Solar Charging-Station Design



Every stage is reproducible from one command (run_analysis.py); all assumptions traceable in config.py

The result: robustness pays



-60%

worst-case annual cost vs the naive design

87% -> 99%

guaranteed fleet service, worst demand

£21,045/yr

robust worst-case cost (down from £52,619)

A small expected-cost premium buys large protection against the demand tail.

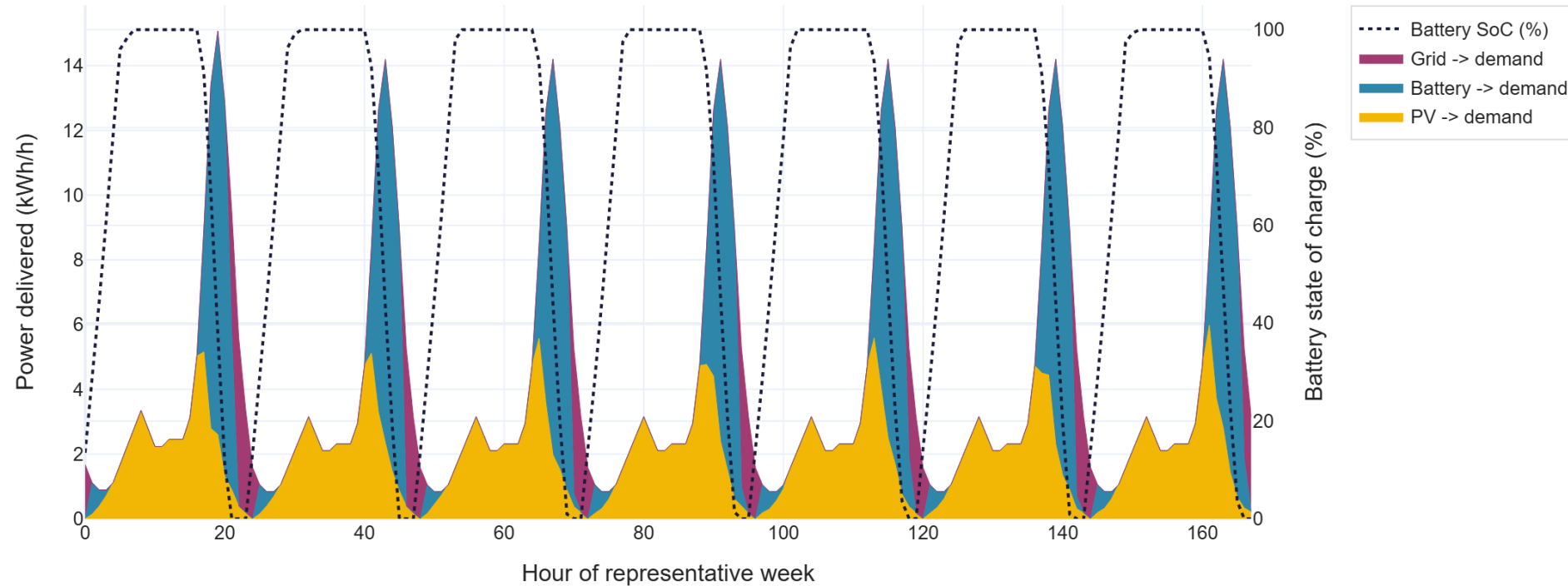
Recommended robust station — Theme 2 in action

25 kWp solar PV

50 kWh battery

4 charge points

Hourly energy-balance dispatch & battery profile (Theme 2)



Battery charges overnight and at midday, then discharges through the evening peak the capped grid cannot meet.

Feasible today — and sustainable

~£68,400

capital cost — off-the-shelf PV, Li-ion & AC chargers

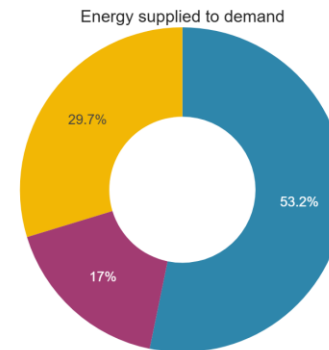
-84% CO₂

vs grid-only charging (6.2 tCO₂/yr saved)

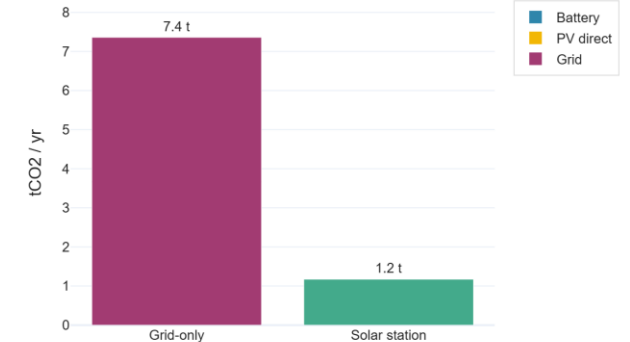
1 command

regenerates all data, figures & report — 8/8 tests pass

Energy mix & carbon savings (Theme 3 sustainability)



Annual CO₂: station vs grid-only



Original, authentic, reproducible

Every line of code and every figure is original, written from scratch. The only external data is the DfT e-scooter dataset, reused under the Open Government Licence; solar and carbon series are physically modelled, not copied. Fixed seed, one-command rebuild, passing tests — built for the AI and plagiarism checks.



**We turn demand uncertainty from a risk
into a design input — and show that robustness pays.**

Thank you.